

COMP3052 Computer Security

Coursework 01 (20%)

Lab Report – Passwords

Learning Aims:

This coursework is designed to enable you to reflect on and report on your computer security experiences in lab sessions. You are expected to design and justify a password and authentication policy based on practical experiments, supported by credible references and personal reflection.

Explanation

You are required to write a detailed report of approximately 1000 words that covers password security as encountered in labs and lectures. Marks will be awarded based on:

- Correctness and completeness of your answers
- Depth of exploration and technical accuracy
- Quality of justification and evidence
- Integration of personal reflection and practical experience

Task Details

A system administrator has asked you to design a new password and authentication policy for their network. You must:

- List and describe your proposed password policy rules (e.g., length, complexity, expiration, reuse, etc.).
- Objectively justify each policy element with evidence from:
 - Your own lab experiments (include screenshots in the appendix — do not reuse provided lab materials)
 - Reputable academic or industry sources (e.g., journals, whitepapers, etc.)
- Reflect subjectively on your lab experiences:
 - How did your experiments inform your policy choices?
 - What challenges or insights did you encounter?
 - How would you improve or adjust your approach in the future?
- Consider additional authentication measures where relevant, and justify their use.
- Explain what attacks your policy aims to prevent.
- Conduct and document additional experimental research beyond the guided lab sessions. This may include testing password strength tools or simulation of attack scenarios.

Formatting & Submission Requirements

- File name: [YourStudentID]_report.pdf
- Cover page must include:
 - Your name
 - Student ID
 - Module code (COMP3052)
 - Date
- Word Count: 900 – 1100 words (excluding references and appendices).
- Submit via Moodle by **6 pm, Monday, April 6, 2026.**

Marking Criteria

Your report will be assessed across four criteria:

- Basic Information & Formatting (10%)
 - Correct file naming, cover page details, structure, and adherence to submission guidelines.
- Listing & Description of Password Policy (10%)
 - Clear, comprehensive, and technically sound policy rules.
- Objective Justification of Policy Elements (30%)
 - Supported by evidence from lab experiments, including relevant screenshots, and reinforced by credible academic or technical references.
- Subjective Justification & Reflection (30%)
 - Demonstrating personal insights and a reflective connection to lab experiences, evidenced by supporting screenshots.
- Additional Experiments (20%)
 - Go beyond basic labs by designing and running own experiments. Examples include creating scripts to evaluate password policies or modeling real-world attack methods. Support findings with screenshots for code and the results.